

Teacher Notes — The Architect's Mind in Orbit

~2 h (105–130 min as written; ~90–100 with optional parts cut) · ages 12–18 · phones + projector + whiteboard · one QR at a time · Full plan:

workshop-en.md

Before the session

- Slides: paste text from [slides-en.md](#) (structure: [slides-outline.md](#)); whiteboard markers; cabin noise on speakers.
- QR PDF — show **one code at a time** on projector ([qr-en.pdf](#)).
- Handouts A3 ([handout-en.pdf](#)); **2 sticky notes per student** (name on each).
- Earphones!** Comms task is spoken audio. iPhone: ring/silent switch → ring. Test one iPhone + one Android the day before.
- Tell students: nothing is uploaded; never type a real name in the app (tap **Skip**).
- Teacher laptop ready to mirror a student's Blueprint. Presets on hub (tap **Presets**): /comms /monitoring /tracking /resource.

Run sheet

Block	You do	Students do
A · Present ~35–45 m	Intro speaker · slide 1 · station history (Salyut→Tiangong) · messy machine-centric design · national-connection slide (ESA/ISS) · Wandt's Destiny-lab photo · "What makes space stressful?" · whiteboard : space → school → home → relax · cognitive link slide · current vs. Guidice future · small steps · 3 tests + MDRS/ISS · age guide (RT 12–13 / MATB 14–15 / N-Back 16–18)	Discuss; contribute to board (stays up all class).
B · Experiment ~45–50 m	QR: RT only — gather spread, confounders, "you vs. you, as a %". Projector : MATB all tasks → each Training page → 2-min mode. Then QR: MATB dry run + walkthrough. QR: Condition Lab — run ① together (speakers off), then ② together (speakers on). MATB opens in a new tab ; students come back and type Final Score under ①/②; read the "% change" sentence. Whiteboard: learning curve, plateau, athlete analogy, order flaw, class tally. Full run card below .	RT one round · MATB practice · Condition Lab · fill handout \$1 .
C · Design ~20–30 m	QR: Simulator — illustrative model, find the slider that moves Overall most. QR: Blueprint — brief + trade-offs; repeat  corridor rule; scenario starts at 45, target 70. Rebuild a high scorer's grid on the laptop and project it. Optional : Model Lab (cut if long; Reset to example afterwards) · A3 group sketch (handout \$3).	Simulator → Blueprint (Mission Success ≥ 70) · handout \$2.
D · Exit ~7 m	Collect two post-its per student. Closing slide : Space for everyone · Architecture = people · Design = trade-offs · You = future of exploration. Thanks + links.	Post-it 1: helps cognition · Post-it 2: hurts cognition (name on each).

Run card — Block B (≈ 45–50 min)

Time	Show	Say / do	Look for
0–8	Slide 15 · QR RT	"Tap the red dot, 8 times." Collect 5 averages aloud. Confounders → "you vs you, as a %".	Averages 200–400 ms. Under 150 = early taps.
8–18	Projector: Presets → Quick Test, then 4 Training pages	Full MATB 30 s → each task 2 min with the phone gesture → full MATB 20 s. No student QR .	Students can name all four tasks. If not, redo the one they can't.
18–26	QR 2min	"Tap I Understand; don't Quit; read Final Score; Skip the name box; write it in <i>Practice</i> ."	Silent phones (mute switch) · stuck on results page · someone quit early → run again.
26–39	QR Condition Lab	① together, speakers off → back to the tab → type score. ② together, speakers on → type score. Read the % sentence.	Both boxes filled → Comparison chart appears. Lost tab → browser tabs icon.
39–47	Whiteboard	Sports curve → plateau → "did ① beat Practice?" → order flaw → tally ↓ ≈ ↑.	Most hands in ↓ or ≈; if ↑ dominates, practice won — that's the point.

Key messages

- Today's ISS is still messy, loud, cramped — design must improve with evidence.
- Scientists measure, compare conditions, account for learning & confounders.
- Blueprint trade-offs: crew mind vs. mission science/engineering.

Closing slide (after exit ticket)

- Space is for everyone.**
- Architecture is about people.**
- Design is a series of trade-offs.**
- You are the future of space exploration.**

Pilot learnings pilot: 60 students, 3 groups of ~20

- Phones mostly OK — try **Chrome** if Safari glitches.
- MATB: demo on projector first; pair if needed; mixed scores normal.
- Blueprint additive — repeat entrance corridor rule; give a 2-minute warning. See [workshop-feedback.md](#).

Expected values (so you can spot a broken phone)

- **Reaction Time:** 8 red dots, no time limit (~1 min). Read **Average Reaction Time**. Typical 200–400 ms on a phone; under ~150 ms = tapped too early.
- **MATB 2-min:** Final Score = Health summed every second; **max 12,000**. Near-perfect $\approx 11,000+$; a struggling first run can be 5,000–8,000. A 5 % drop is e.g. 10,000 $\rightarrow$ 9,500.
- **Condition Lab:** % change = $(\textcircled{2} - \textcircled{1}) \div \textcircled{1}$. Swings of ± 10 % are normal for one run each; the class tally is the real result.
- **Simulator:** default sliders $\rightarrow$ Overall ≈ 61 ; Noise to 100 $\rightarrow \approx 40$; everything worst $\rightarrow 0$; everything best $\rightarrow 100$.
- **Blueprint:** starting scenario = 45. Checklist items alone = 50. Keep 1 life support + 1 science rack + bike, add 2 stowage, sleeping quarters, 2 acoustic zones, plant wall, circadian light, med bay, greenhouse $\rightarrow 70$. Realistic ceiling ≈ 80 ; stretch goal ≥ 78 .

Troubleshooting

- **Running long?** Cut Model Lab $\rightarrow$ A3 sketch $\rightarrow$ shorten history.
- **Too many QRs?** One on screen at a time only.
- **No room speakers?** Use the in-app cabin noise in the Condition Lab on each phone (earphones), at the default volume.
- **No radio voice?** Phone muted (iPhone side switch) or volume down. Still nothing $\rightarrow$ Chrome.
- **"New High Score" box asks for a name?** Tap **Skip** — it appears on every fresh phone. Never type real names; nothing is uploaded, but keep it clean.
- **Which number is the score? Final Score** at the top of the *Simulation Complete* screen (max 12,000 = Health 100 for 120 s).
- **Wrong language?** [?lng=sv](#) or [?lng=el](#)

Links

- /reaction-default · Presets: /comms /monitoring /tracking /resource · /2min · /condition-lab
- /simulator · /blueprint · /model-lab (optional)

Sources: Comstock & Arnegard, *MATB*, NASA TM-104174 (1992) · Dinges & Powell, *PVT*, Behav. Res. Methods (1985) · Szalma & Hancock, noise & performance meta-analysis, Psychol. Bull. (2011) · Clark, *Acoustic Issues in Human Spaceflight*, NASA (2001) · ESA blog "I need more space" (2024) on the Orbital Architecture ISS experiment. Full list with links: [workshop-en.md](#) $\rightarrow$ References & further reading.